HDOC-50

Majority Element in a Variable-Length Integer Array Using Pointers

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. Pass the array to a function named majority_element() using its address. Use pointers to pass the array between the calling function and majority_element(). The function must count the occurrences of each distinct element, find the element that appears strictly more than size / 2 times, display the majority element if it exists, and display -1 if no majority element exists.

Algorithm

1. Read the size of the array, n.
2. Declare a variable-length integer array nums[n] and read its elements.
3. Pass the base address of nums[] and n to majority_element().
4. For every distinct element, count its occurrences using pointer arithmetic.
5. If any element occurs more than n / 2 times, display that element and return.
6. If no such element is found, display -1.

C Program

```
#include <stdio.h>

void majority_element(int *nums, int size)
{
    int i, j, count;

    for (i = 0; i < size; i++)
    {
        /* Skip an element if it has already been counted */
        int alreadyCounted = 0;
        for (j = 0; j < i; j++)
        {
            if (*(nums + i) == *(nums + j))
            {
                alreadyCounted = 1;
                break;
            }
        }

        if (alreadyCounted)
            continue;

        count = 0;
        for (j = 0; j < size; j++)
        {
            if (*(nums + i) == *(nums + j))
                count++;
        }

        if (count > size / 2)
        {
            printf("Majority element: %d\n", *(nums + i));
            return;
        }
    }

    printf("Majority element: -1\n");
}

int main(void)
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);
```

```
    if (n <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    int nums[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", nums + i);

    majority_element(nums, n);

    return 0;
}
```

Test Cases

Test Case	Input	Expected Output	Result
1	7 2 2 1 2 3 2 2	Majority element: 2	Pass
2	5 1 2 3 2 1	Majority element: -1	Pass
3	6 4 4 4 4 2 3	Majority element: 4	Pass

Result

The program successfully uses pointers to pass a variable-length integer array to majority_element(), counts occurrences of distinct elements, and displays the majority element when its count is strictly greater than size / 2; otherwise, it displays -1.